

Experiment 13

Question

Perform Perspective Transformation on the Video.

Aim

To read a video using Python and OpenCV and apply perspective transformation to each frame of the video.

Procedure

1. Import the OpenCV (cv2) and NumPy (numpy) libraries.
2. Read the input video using cv2.VideoCapture().
3. Read the video frame by frame.
4. Define four source points and four destination points.
5. Compute the perspective transformation matrix.
6. Apply the perspective transformation to each frame.
7. Display the original and transformed video.

Code

```
import cv2
```

```
import numpy as np
```

```
cap = cv2.VideoCapture("sample.mp4")
```

```
while True:

    ret, frame = cap.read()

    if not ret:
        break

    frame = cv2.resize(frame, (640,480))

    p1 = np.float32([[100,100],[500,100],[100,400],[500,400]])
    p2 = np.float32([[0,0],[640,0],[0,480],[640,480]])

    M = cv2.getPerspectiveTransform(p1, p2)
    out = cv2.warpPerspective(frame, M, (640,480))

    cv2.imshow("Original", frame)
    cv2.imshow("Output", out)

    if cv2.waitKey(30) == ord('q'):
        break

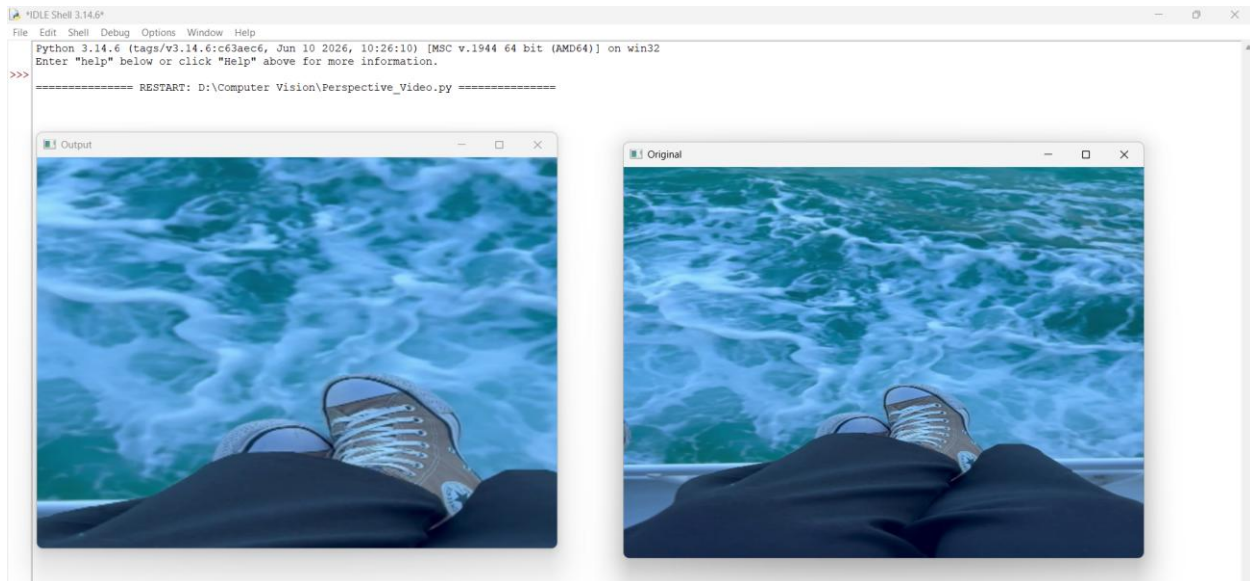
cap.release()
cv2.destroyAllWindows()
```

Input

Input Video: sample.mp4

Output

- Perspective Transformed Video



Result

The perspective transformation was successfully applied to each frame of the input video using OpenCV. The transformed video was displayed successfully.